Exercise on Committee Selection

- Suppose there are 40 faculty members in the computer science department and 30 in the mathematics department. How many ways are there to select a committee that has 4 faculty members from the computer science and 3 from the mathematics department?
- **Solution**
- There are $C(40,4)$ ways to select the 4 computer science faculty members.
- For each fixed set of selected CSE faculty members, there are $C(30,3)$ ways to select the 3 Math faculty members.
- By the product rule so there are a total of $C(40,4) \times C(30,3) = \frac{40!}{4!36!}\frac{30!}{3!27!}$ possible ways.